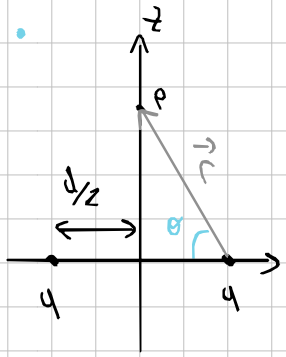
1.



$P(0,z)$ $\vec{E}$ in P!

$\vec{E}_{TOT} = \vec{E}_z + \vec{E}_z$

$\vec{E}_y = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{r^2} \sin(\theta)\, \hat{k}$

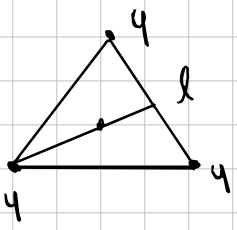
$\vec{E}_y = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{r^2} \cdot \frac{z}{r}\, \hat{j}$ $\qquad r = \sqrt{\frac{d^2}{4} + z^2}$

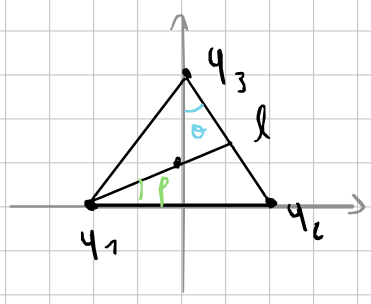
$\Rightarrow \vec{E}_y = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q\, z}{\left(\frac{d^2}{4} + z^2\right)^{3/2}}\, \hat{k} \quad \Rightarrow \quad \vec{E}_{TOT} = \frac{q \cdot z}{2\pi\varepsilon_0} \cdot \frac{1}{\left(\frac{d^2}{4} + z^2\right)^{3/2}}$

Dass $d/2 \to 0$ $\quad (z \to \infty)$ $\quad \vec{E}_{TOT} = \frac{q \cdot z}{2\pi\varepsilon_0} \cdot \frac{1}{z^3 \left(1 + \frac{d^2}{4z^2}\right)^{3/2}}\, \hat{k}$

$\Rightarrow$ pcr $d/2 \to 0$ $\quad \vec{E}_{TOT} = \frac{q}{2\pi\varepsilon_0} \cdot \frac{1}{z^2}\, \hat{k}$

2.





$\vec{F}_{42} = \vec{F}_{41} + \vec{F}_{43} = q \cdot \vec{E}_1 + q \cdot \vec{E}_2$ $\qquad \vec{E}_1 = < \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{l^2}, 0, 0 >$

$\vec{E}_2 = < \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{l^2} \cdot \sin(\theta), \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{l^2} \cdot \cos(\theta), 0 >$

$\Rightarrow \vec{E}_2 = < \frac{q}{4\pi\varepsilon_0} \cdot \frac{1}{l^2} \cdot \left(\frac{l}{2}\right) \cdot \frac{1}{l}, \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{l^2} \cdot \left(\frac{\sqrt{3}}{2}\right) l \cdot \frac{1}{l}, 0 >$

$\vec{E}_1 + \vec{E}_2 = < \frac{q}{4\pi\varepsilon_0} \cdot \frac{1}{l^2} \cdot \left(1 + \frac{1}{2}\right), \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{l^2} \cdot \frac{\sqrt{3}}{2}, 0 >$

$\|\vec{E}_{TOT}\| = \frac{q}{4\pi\varepsilon_0} \cdot \frac{1}{l^2} \cdot \sqrt{\frac{q}{4} + \frac{3}{4}} = \frac{q}{4\pi\varepsilon_0} \cdot \frac{1}{l^2} \cdot \sqrt{3} \Rightarrow \|\vec{F}\| = \vec{E}_{TOT} \cdot q = \frac{\sqrt{3}}{4\pi\varepsilon_0} \cdot \frac{q^2}{l^2}$

$-\vec{E}$ al centro $\qquad E_{y1}$ e E_{y2} si annullano sulle x. E_{y3} non ha componente sulle x. $\qquad \vec{E}_{TOT} = \vec{E}_{y1} + \vec{E}_{y2} - \vec{E}_{y3}$

$\vec{E}_{y1} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{i} \cdot \sin(\beta) = \frac{1}{4\pi\varepsilon_0} \cdot \frac{1}{\frac{l^2}{4}} \cdot \left(\frac{l}{2\sqrt{3}}\right) \cdot \frac{1}{i} \cdot q$

$i = \sqrt{\left(\frac{l}{2\sqrt{3}}\right)^2 + \left(\frac{l}{2}\right)^2} = \sqrt{\frac{l^2 + 3l^2}{12}} = \frac{l}{\sqrt{3}}$

$\Rightarrow E_{y1} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{1}{\frac{l^2}{3}} \cdot \frac{l}{2\sqrt{3}} \cdot \frac{1}{\frac{l}{\sqrt{3}}} = \frac{3q}{8\pi\varepsilon_0 l^2}$ $\qquad E_{y2} = E_{y1}$

distanza q_3 – baricentro $\Rightarrow \frac{\sqrt{3}}{2} l - \frac{l}{2\sqrt{3}} = \frac{2l}{2\sqrt{3}} = \frac{l}{\sqrt{3}}$

↑ altezza

$\Rightarrow E_{y3} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{\frac{l^2}{3}} \cdot 1$

$\Rightarrow E_{y3} = \frac{3}{4} \frac{q}{\pi\varepsilon_0 l^2}$

$E_{y1} + E_{y2} - E_{y3} = 0$

$U = \frac{1}{2} \sum_i q_i V_i(r_i)$

Potenziale nel punto in cui è q_1 = potenziale di q_2 + potenziale di q_3

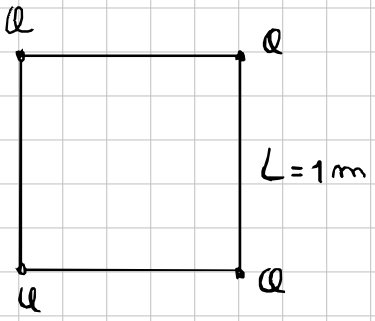
$= \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{l} + \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{l} = \frac{1}{2\pi\varepsilon_0} \cdot \frac{q}{l}$

↑ distanza da q_1

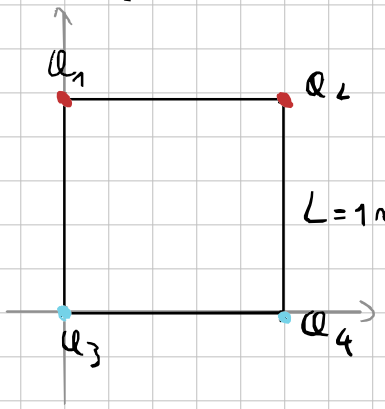
Uguale per le altre due cariche

$U = \frac{1}{2} \cdot 3 \cdot \frac{q}{4\pi\varepsilon_0} \cdot \frac{1}{l} = \frac{3}{4} \frac{q}{\pi\varepsilon_0 l}$

3.



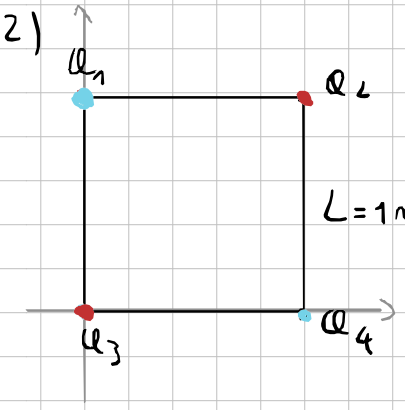
1)



$$U = \frac{1}{2} \sum_i q_i V_i(r_i) \qquad U_{sistema} = \sum U_{coppia}$$

$$U = \left[\frac{kQ^2}{L} - \frac{kQ^2}{L} + \frac{kQ^2}{L} - \frac{kQ^2}{L} - \frac{kQ^2}{\sqrt{2}L} - \frac{kQ^2}{\sqrt{2}L} \right] = \left[-\sqrt{2}\,\frac{kQ^2}{L} \right]$$

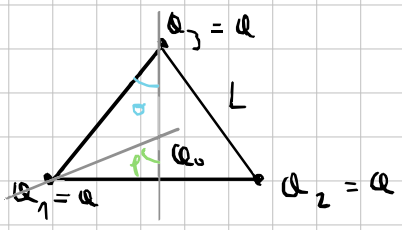
2)



probabilmente sbagliato

$$U = \left[\frac{kQ^2}{L} - \frac{kQ^2}{L} - \frac{kQ^2}{L} - \frac{kQ^2}{L} + \frac{kQ^2}{\sqrt{2}L} + \frac{kQ^2}{\sqrt{2}L} \right] = \left[-\frac{4}{L}kQ^2 + \frac{\sqrt{2}}{L}kQ^2 \right]$$

4.



$Q = 2{,}2 \cdot 10^{-5}\ C$

$L = 0{,}2\ M$

Ti cmi: sempre avvio dalla geometria del sistema

$$\vec{F} = \frac{1}{4\pi\varepsilon_0} \cdot \frac{Q_0 Q}{r^2}\,\hat{r}$$

$$|\vec{F}| \text{ su } q_1 = |\vec{F}_{q_2 q_1}| + |\vec{F}_{q_3 q_1}|$$

$$E_{q_3 q_1} = \left\langle \frac{k\,q}{L^2} \cdot \sin(\alpha), \frac{kq}{L^2}\cos(\alpha), 0 \right\rangle = \left\langle \frac{kq}{L^2}\left(\frac{L}{2}\right)\frac{1}{L}, \frac{kq}{L^2}\cdot\frac{\sqrt{3}}{2}L\,\frac{1}{L}, 0 \right\rangle$$

$$= \left\langle \frac{kq}{2L^2}, \frac{kq}{2L^2}\cdot\sqrt{3}, 0 \right\rangle$$

$$E_{q_2 q_1} = \left\langle \frac{kq}{L^2}, 0, 0 \right\rangle$$

$$\vec{F} = q\cdot(\vec{E}_1 + \vec{E}_2) = q\left\langle \frac{3}{2}\frac{kq}{L^2}, \frac{\sqrt{3}}{2}\frac{kq}{L^2}, 0 \right\rangle$$

$$|\vec{F}| = q\cdot\sqrt{\frac{k^2q^2}{L^4}\left(\frac{4}{4}+\frac{3}{4}\right)} = \sqrt{3}\,\frac{kQ^2}{L^2}$$

Altezza baricentro $= \frac{L}{2\sqrt{3}}$ ipotenusa $= \frac{L}{\sqrt{3}}$

$$F_{q_0 q_1} = q_1 \cdot \vec{E}_{q_0 q_1} \qquad \vec{E}_{q_0 q_1} = \left\langle \frac{1}{4\pi\varepsilon_0}\cdot\frac{Q_0}{\frac{L^2}{3}}\cdot\sin(\beta), k\frac{Q_0}{\frac{L^2}{3}}\cos(\beta), 0 \right\rangle$$

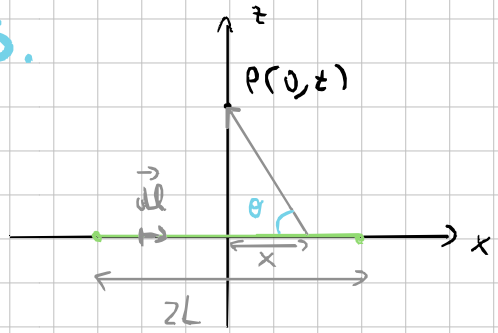
$$= \left\langle \frac{3kQ_0}{L^2}\cdot\left(\frac{L}{2}\right)\cdot\left(\frac{\sqrt{3}}{L}\right), \frac{3kQ_0}{L^2}\cdot\left(\frac{L}{2\sqrt{3}}\right)\frac{\sqrt{3}}{L}, 0 \right\rangle$$

$$|\vec{E}| = \frac{kQ_0}{L^2}\sqrt{\left(\frac{3\sqrt{3}}{2}\right)^2 + \left(\frac{3}{2}\right)^2} = \frac{kQ_0}{L^2}\sqrt{\frac{27}{4}+\frac{9}{4}} = \frac{3kQ_0}{L^2} \qquad |\vec{F}| = \frac{3kQ_0Q}{L^2}$$

Affinché $|\vec{F}| = 0$ $\quad \frac{\sqrt{3}\,kQ^2}{L^2} + \frac{3kQ_0Q}{L^2} = 0 \quad \Rightarrow \frac{\sqrt{3}}{L^2}kQ^2 = -\frac{3kQ_0Q}{L^2}$

$$\Rightarrow Q_0 = -\frac{\sqrt{3}}{3}Q$$

5.



Q, $\lambda = \frac{Q}{2L}$ $\quad \vec{E} = ?$

$\vec{E} = E_z \hat{j}$

$dE_z = \frac{1}{4\pi\varepsilon_0} \cdot \frac{dQ}{r^2} \cdot \sin(\theta)$

$r = \sqrt{x^2+z^2}$ $\quad \sin(\theta) = \frac{z}{r}$

$\Rightarrow dE_z = \frac{1}{4\pi\varepsilon_0} \cdot \frac{dQ}{(x^2+z^2)^{3/2}} \cdot z \cdot \hat{k}$

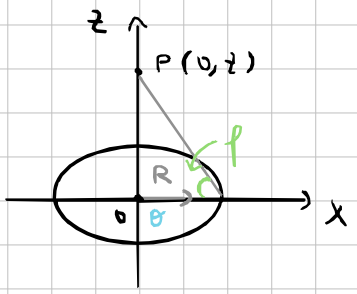
$\vec{E}_z = \int dE_z \, dl$ $\quad dQ = \lambda \cdot dl$ $\quad \Rightarrow \vec{E}_z = \int \frac{k \lambda \, dl}{(x^2+z^2)^{3/2}} z$

$\vec{dl} = \langle dx, 0, 0 \rangle$ $\quad \Rightarrow \vec{E}_z = k\lambda z \int_{-L}^{L} \frac{1}{(x^2+z^2)^{3/2}} \cdot dx$

$\Rightarrow \vec{E}_z = k\lambda z \left[\frac{1}{z^2} \cdot \frac{x}{\sqrt{x^2+z^2}} \right]_{-L}^{L} = \frac{1}{4\pi\varepsilon_0} \lambda \cdot \frac{1}{z} \left[\frac{L}{\sqrt{L^2+z^2}} + \frac{L}{\sqrt{L^2+z^2}} \right]$

$= \frac{1}{4\pi\varepsilon_0} \lambda \cdot \frac{1}{z} \cdot \frac{2L}{\sqrt{L^2+z^2}} = \frac{\lambda L}{2\pi\varepsilon_0 \sqrt{L^2+z^2}}$

6.



Anello con densità λ e carica Q

$\vec{E} = ?$

$\vec{E} = E_z \hat{k}$

$$d\vec{E}_z = \frac{1}{4\pi\varepsilon_0} \cdot \frac{dQ}{r^2} \cdot \sin(\varphi)\ \hat{k} \qquad r^2 = \sqrt{R^2 + z^2}$$

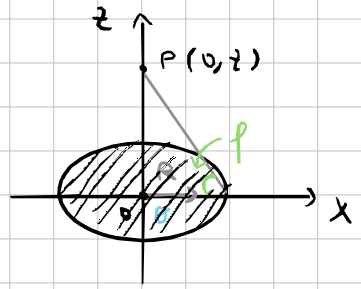
$$\sin(\varphi) = \frac{z}{\sqrt{R^2+z^2}}$$

$dQ = \lambda \cdot dl \qquad dl = R \cdot d\theta$

$$E_z = \int_0^{2\pi} \frac{1}{4\pi\varepsilon_0} \frac{z}{(R^2+z^2)^{3/2}} \cdot \lambda \cdot R \cdot d\theta = \frac{1}{4\pi\varepsilon_0} \cdot \frac{\lambda R z}{(R^2+z^2)^{3/2}} \cdot \int_0^{2\pi} d\theta$$

$$= \frac{\lambda R z}{2(R^2+z^2)^{3/2}\varepsilon_0} \qquad Q = \lambda \cdot 2\pi R \Rightarrow \lambda = \frac{Q}{2\pi R} \Rightarrow \frac{Q z}{4\pi\varepsilon_0 (R^2+z^2)^{3/2}}\ \hat{k}$$

7.



Disco σ densità di carica superficiale costante

$\vec{E} = ?$

$\sigma = \frac{Q}{A} \Rightarrow \sigma = \frac{Q}{\pi R^2}$

VARIABILE!!

$\vec{E} = E_z \hat{k}$ $\quad dE_z = \frac{1}{4\pi\varepsilon_0} \cdot \frac{dQ}{r^2} \cos(\theta)$ $\quad r = \sqrt{r^2+z^2}$ $\quad \cos(\theta) = \frac{z}{\sqrt{r^2+z^2}}$

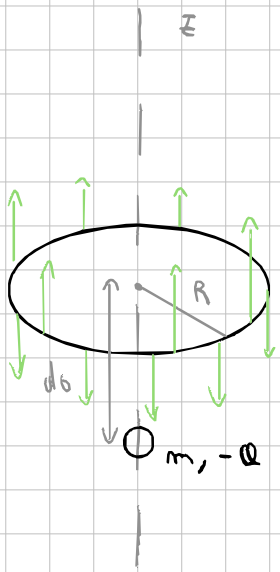
$dQ = \sigma \cdot 2\pi r\, dr$ ← r variabile per ogni dQ!

$E_z = \int \frac{1}{4\pi\varepsilon_0} \cdot \frac{\sigma \cdot 2\pi r\, dr \cdot z}{(r^2+z^2)^{3/2}} = \frac{\sigma}{2\varepsilon_0} \cdot \int_0^R \frac{r}{(r^2+z^2)^{3/2}}\, dr = \frac{\sigma}{2\varepsilon_0} \frac{1}{2} \int_0^R 2r\,(r^2+z^2)^{-3/2}$

$= \frac{\sigma}{2\varepsilon_0} \cdot \frac{1}{2} \left[-2 \cdot \frac{1}{(r^2+z^2)^{1/2}} \right]_0^R = \frac{\sigma z}{2\varepsilon_0} \left[-\frac{1}{\sqrt{R^2+z^2}} + \frac{1}{z} \right]$ ← $\frac{1}{-\frac{1}{2}}$

$\Rightarrow \vec{E} = \frac{\sigma}{2\varepsilon_0} \left[1 - \frac{z}{\sqrt{R^2+z^2}} \right]$

23.



Anello isolante raggio R, carica Q.

Particella di massa m e carica -Q sull'asse dell'anello a distanza d_0 con quale velocità trasmessa nel centro dell'anello? in che punto si ferma?

Per principio di conservazione dell'energia meccanica, l'energia meccanica totale, in presenza di forze conservative, si conserva. L'energia meccanica totale è data da $U + K$ con K= en. cinetica e U en. potenziale (in questo caso energia elettrostatica). Per principio di conservazione dell'energia meccanica

$U_{INIZIALE} + K_{INIZIALE} = U_{FINALE} + K_{FINALE}$

U_i (della particella rispetto all'anello) $= \frac{1}{4\pi\varepsilon_0} \cdot \frac{Q^2}{\sqrt{R^2 + d_0^2}}$

$K_i = \frac{1}{2} m v_i^2 \quad v_i = 0 \Rightarrow K_i = 0$

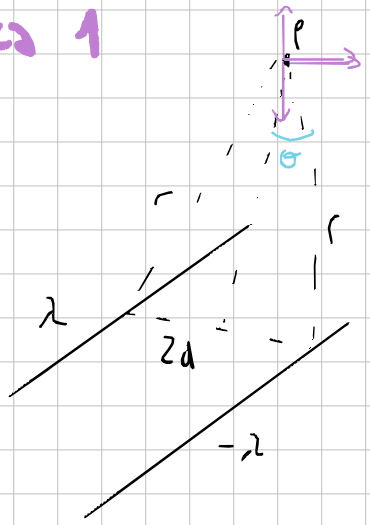
U_{FINALE} = U quando la particella è al centro dell'anello $= \frac{1}{4\pi\varepsilon_0} \cdot \frac{Q^2}{\sqrt{R^2}}$

$K_f = \frac{1}{2} m v_f^2$

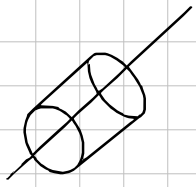
$$\Rightarrow \frac{1}{4\pi\varepsilon_0} \cdot \frac{Q^2}{\sqrt{R^2 + d_0^2}} + 0 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{Q^2}{R} + \frac{1}{2} m v_f^2$$

$$\Rightarrow v_f = \sqrt{\frac{Q^2}{2\pi\varepsilon_0 m} \cdot \left(\frac{1}{R} - \frac{1}{\sqrt{R^2 + d_0^2}} \right)}$$

1 Verifica 1



- $\vec{E}$ piano con $-\lambda$



$$\Phi(\vec{E}) = \frac{Q}{\varepsilon_0} \qquad Q = \lambda \cdot L$$

$$\Phi(\vec{E}) = \vec{E} \cdot 2\pi R L$$

$$\vec{E} = \frac{\lambda L}{2\pi R L} \cdot \frac{1}{\varepsilon_0} = -\frac{\lambda}{2\pi r \varepsilon_0}\,\hat{i}$$

- $\vec{E}$ in P

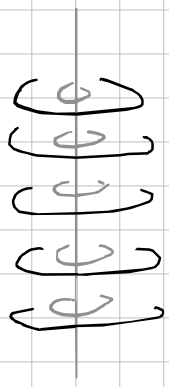
$$\vec{E} = E_x\,\hat{i} \qquad E_x = \frac{-\lambda}{2\pi r \varepsilon_0} \cdot \sin(\theta)\,\hat{i} \qquad \sin(\theta) = \frac{d}{r}$$

$$\Rightarrow E_x = -\frac{\lambda}{2\pi r \varepsilon_0} \cdot \frac{d}{r} \qquad \vec{E} = 2 \cdot E_x = \frac{\lambda}{\pi \varepsilon_0} \cdot \frac{d}{r^2}\,\hat{i}$$

- ΔV tra P e la sua proiezione sul piano dei fili

$\vec{E}$ è diretto come l'asse x e lo spostamento da P a O (su cui calcolare il potenziale) è sull'asse y (dell'asse y) $\Rightarrow \Delta V = 0$

Esercizio verifica



R_1 raggio esterno e densità di spire n_1, $i_1 = i_0 \cdot \cos(\omega t + \phi)$

R_2 raggio interno e densità di spire n_2, $i_2 = i_0 \cdot \sin(\omega t)$

$\vec{B}$ nello spazio ?

$\vec{E}$ nel solenoide più piccolo ?

① dentro al solenoide più piccolo

② tra i due solenoidi

③ Esterno

- Per l. di Biot-Savart nel solenoide più piccolo $B = \mu_0 \cdot I \cdot n$

$\Rightarrow B = \mu_0 n_1 \cdot i_0 \cdot \cos(\omega t + \phi)$

- In quello più grande $B = \mu_0 n_2 \cdot i_0 \cdot \sin(\omega t)$

Dunque:

① $\vec{B} = \mu_0 i_0 \cdot (n_1 \cos(\omega t + \phi) + n_2 \sin(\omega t))$

② $\vec{B} = \mu_0 i_0 \sin(\omega t) \cdot n_2$

③ $\vec{B} = 0$

Le correnti variano nel tempo $\Rightarrow \vec{B}$ varia $\Rightarrow$ si ha un campo elettrico indotto.

Per l. di Faraday-Neumann-Lenz

$$f_{ind} = -\frac{\Phi(\vec{B})}{dt} = \oint \vec{E}_{IND} \cdot \vec{dl}$$

$\Phi(\vec{B})$ = flusso attraverso la superficie circoscritta dal solenoide più piccolo

$$\Rightarrow \Phi(\vec{B}) = \int_\Sigma \vec{B} \cdot \vec{d\Sigma} = B \cdot \int_\Sigma d\Sigma = B \cdot \pi R_2^2$$

$$= \mu_0 i_0 \cdot (n_1 \cos(\omega t + \phi) + n_2 \sin(\omega t)) \cdot \pi R_2^2$$

$$\frac{\partial}{\partial t}(\Phi(\vec{B})) = \mu_0 i_0 \left(-n_1 \omega \sin(\omega t + \phi) + n_2 \omega \sin(\omega t)\right) \cdot \pi R_2^2$$

$$\oint \vec{E}_{IND} \cdot \vec{dl} = \mu_0 i_0 \left(-n_1 \omega \sin(\omega t + \phi) + n_2 \omega \sin(\omega t)\right) \cdot \pi R_2^2$$

$$\vec{E} \cdot 2\pi R_2 = \mu_0 i_0 \left(-n_1 \omega \sin(\omega t + \phi) + n_2 \omega \sin(\omega t)\right) \cdot \pi R_2^2$$

↑ circuitazione su πR_2^2

$$\vec{E} = \frac{\mu_0 i_0 \left(-n_1 \omega \sin(\omega t + \phi) + n_2 \omega \sin(\omega t)\right) \cdot \pi R_2^2}{2\pi R_2}$$

$$\Rightarrow \vec{E} = \frac{1}{2}\left(\mu_0 i_0 R_2 \left(-n_1 \omega \sin(\omega t + \phi) + n_2 \omega \sin(\omega t)\right)\right)$$

In realtà andrebbe calcolato attraverso una generica r, non su R_2, dunque

$\oint \vec{E}_{IND} \, dl = \vec{E} \cdot 2\pi r$ e $\Phi(\vec{B}) = B \cdot \pi r^2$

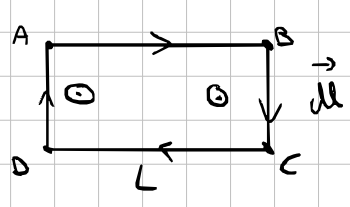
$$\Rightarrow \vec{E} = \frac{1}{2}\left(\mu_0 i_0 r \left(-n_1 \omega \sin(\omega t + \phi) + n_2 \omega \sin(\omega t)\right)\right)$$

$$\oint \vec{B} \cdot dl = \mu_0 \sum_i I_i^{concat}$$

$$\mu_0 \sum_i I_i^{concat} = \mu_0 \cdot I \cdot n \cdot L$$

$$\oint \vec{B} \cdot dl = \int_A^B \vec{B} \cdot \vec{dl} + \int_B^C \vec{B} \cdot \vec{dl} + \int_C^D \vec{B} \cdot \vec{dl} + \int_D^A \vec{B} \cdot \vec{dl}$$

$= B \cdot L$; $= 0$ * ; $= 0$; $= 0$ *



* 0 perché $\vec{dl} \perp BC$ e a AD

$$\Rightarrow B \cdot L = \mu_0 I \, n \, L \quad \Rightarrow B = \mu_0 I n$$

Nota: assumiamo sempre solenoidi ideali (lunghezza $\gg$ diametro) per cui il campo esterno è molto debole e dunque trascurabile